Sarthak Singh

LinkedIn: linkedin/sarthak35
GitHub:github/SarthakSingh5
Email:sarthak3596@gmail.com
Mobile:+91-9306956061

SKILLS

- **Languages**: C++, C, C#, Java, Python, Javascript, PHP, SQL,GDScript
  Frameworks: HTML and CSS, Bootstrap, Tailwind, NodeJS, React, Express
- **Tools/Platforms**: MySQL, MongoDB, Blender,Unity,Godot
- **Database**: Mongoose, MySQL
- **Technical Skills**: Data Structures and Algorithms (DSA), Object Oriented Programming (OOP)
- **Foundations**: DBMS, Operating Systems, Computer Networking, Software Engineering Principles
- **Soft Skills**: Problem-Solving , Adaptability , Creativity

PROJECTS

Apradh – Turn Based Strategy Shooting Game | GitHub — Jan' 25

- Designed and developed an turn-based crime-strategy game inspired by North Indian,Pakistan socio-cultural settings.
- Implemented advanced AI systems and territorial dominance mechanics in campaign map(strategy) and battle maps(shooting).
- Built multiple factions(facial features,clothing,hair), districts, and culturally grounded locations to enhance immersion and world-building consistency.
- Created custom scripts for AI cover movement, high/low cover logic, peeking behavior, and scene-based combat interactions.

Technologies: Unity, C#, NavMesh, Coroutines, Terrain System, Animation Rigging, Blender

CMS – Full-Stack Content Management System| GitHub — Feb' 26

- Developed a comprehensive Content Management System (CMS) using the MERN stack to streamline academic administration and enrollment.
- Engineered secure RESTful APIs and database schemas to manage data relationships between users, courses, and instructors.
- Implemented role-based access control and authentication using JWT to ensure data integrity and personalized dashboards.
- Optimized front-end responsiveness and state management to provide a seamless interface for real-time data updates.

Technologies: MongoDB, Express.js, React, Node.js, JavaScript, Tailwind, REST API

Automated Deadlock Detection Tool | GitHub — Nov' 24

- Developed a specialized utility to identify and visualize potential deadlock conditions in multi-threaded environments.
- Implemented resource-allocation graph algorithms to detect circular wait dependencies and resource contention.
- Created an automated reporting system that logs thread states and provides actionable insights for synchronization optimization.
- Designed a lightweight monitoring engine to track lock acquisition patterns without significant performance overhead.

Technologies: C++, Operating Systems, Multithreading, Graph Algorithms, System Programming

CERTIFICATES

- Full Stack Development – Cipher School **|Link** — July' 25
- Advanced Computer Networks – NPTEL **|Link** — Apr' 25
- Introduction to Hardware and Operating System – IBM **|Link** — Sep' 24
- The Bits and Bytes of Computer Networking – Google **|Link** — Sep' 24

ACHIEVEMENTS

- Solved 250+ coding problems on **LeetCode**, sharpening problem-solving skills. — Feb' 25
- Earned 100 days badges on **LeetCode** — Nov' 25

EDUCATION

Lovely Professional University — Phagwara, Punjab
Bachelor of Technology - Computer Science and Engineering: **CGPA: 7.53** — Aug' 23 – Present

Maharana Pratap Public School — Kurukshetra,Haryana

Intermediate: **Percentage: 82%** Apr' 21 – Mar' 22

Maharana Pratap Public School Kurukshetra,Haryana

Matriculation: **Percentage: 88%** Apr' 19- Mar' 20